

(116) Resume Building Session

Check Telnet SSH Status

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Ask Gemini

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QEMU (Win-3)

Microsoft Windows [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.
C:\Documents and Settings\User>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

 Connection-specific DNS Suffix . :
 IP Address : 172.36.0.10
 Subnet Mask : 255.255.0.0
 Default Gateway : 172.36.0.1

C:\Documents and Settings\User>

!!!!!! N

Win-1

192.50.20.100

Win-2

223.0.50.10

Win-3

172.36.0.10

Physical

01 : 00 : 00

START

Search

11:19 PM
7/19/2026

(116) Resume Building Session

Check Telnet SSH Status

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Ask Gemini

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QEMU (Win-3)

Microsoft Windows [Version 5.1.2600]
Copyright 1985-2001 Microsoft Corp.
C:\Documents and Settings\User>ping 192.50.20.100
Pinging 192.50.20.100 with 32 bytes of data:
Reply from 192.50.20.100: bytes=32 time=2ms TTL=124
Reply from 192.50.20.100: bytes=32 time=2ms TTL=124
Reply from 192.50.20.100: bytes=32 time=1ms TTL=124
Reply from 192.50.20.100: bytes=32 time=3ms TTL=124
Ping statistics for 192.50.20.100:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 3ms, Average = 2ms
C:\Documents and Settings\User>ping 223.0.50.10
Pinging 223.0.50.10 with 32 bytes of data:
Reply from 223.0.50.10: bytes=32 time=1ms TTL=124
Reply from 223.0.50.10: bytes=32 time=2ms TTL=124
Reply from 223.0.50.10: bytes=32 time=2ms TTL=124
Reply from 223.0.50.10: bytes=32 time=2ms TTL=124
Ping statistics for 223.0.50.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 2ms, Average = 1ms
C:\Documents and Settings\User>ping 172.36.0.10
Pinging 172.36.0.10 with 32 bytes of data:
Reply from 172.36.0.10: bytes=32 time<1ms TTL=128
Reply from 172.36.0.10: bytes=32 time<1ms TTL=128
Reply from 172.36.0.10: bytes=32 time<1ms TTL=128
Reply from 172.36.0.10: bytes=32 time<1ms TTL=128
Ping statistics for 172.36.0.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\Documents and Settings\User>

!!!!!! N

Win-1

192.50.20.100

Win-2

223.0.50.10

Win-3

172.36.0.10

R5

Net

Start

Search

99

Physical

01 : 00 : 00

START

12:50 AM 7/20/2026

(116) Resume Building Session

Check Telnet SSH Status

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New folder

Gmail

YouTube

Map

Notifications

!!!!!! N

QEMU (Win-2)

C:\WINDOWS\system32\cmd.exe

Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\User>ping 192.50.20.100

Pinging 192.50.20.100 with 32 bytes of data:

Reply from 192.50.20.100: bytes=32 time=1ms TTL=125

Reply from 192.50.20.100: bytes=32 time=1ms TTL=125

Reply from 192.50.20.100: bytes=32 time=1ms TTL=125

Reply from 192.50.20.100: bytes=32 time=1ms TTL=125

Ping statistics for 192.50.20.100:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\Documents and Settings\User>ping 172.36.0.10

Pinging 172.36.0.10 with 32 bytes of data:

Reply from 172.36.0.10: bytes=32 time=1ms TTL=124

Reply from 172.36.0.10: bytes=32 time=1ms TTL=124

Reply from 172.36.0.10: bytes=32 time=1ms TTL=124

Reply from 172.36.0.10: bytes=32 time=1ms TTL=124

Ping statistics for 172.36.0.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Documents and Settings\User>ping 223.0.50.10

Pinging 223.0.50.10 with 32 bytes of data:

Reply from 223.0.50.10: bytes=32 time<1ms TTL=128

Reply from 223.0.50.10: bytes=32 time<1ms TTL=128

Reply from 223.0.50.10: bytes=32 time<1ms TTL=128

Reply from 223.0.50.10: bytes=32 time<1ms TTL=128

Ping statistics for 223.0.50.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Documents and Settings\User>

192.50.20.100

Win-1

Win-2

223.0.50.10

172.36.0.10

Win-3

Net

Physical

01 : 00 : 00

START

12:48 AM

7/20/2026

(116) Resume Building Session

Check Telnet SSH Status

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New folder

Gmail

YouTube

Map

QEMU (Win-1)

C:\WINDOWS\system32\cmd.exe

Microsoft Windows [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\User>ping 223.0.50.10

Pinging 223.0.50.10 with 32 bytes of data:

Reply from 223.0.50.10: bytes=32 time=2ms TTL=125

Reply from 223.0.50.10: bytes=32 time=2ms TTL=125

Reply from 223.0.50.10: bytes=32 time=1ms TTL=125

Reply from 223.0.50.10: bytes=32 time=1ms TTL=125

Ping statistics for 223.0.50.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Documents and Settings\User>ping 172.36.0.10

Pinging 172.36.0.10 with 32 bytes of data:

Reply from 172.36.0.10: bytes=32 time=3ms TTL=124

Reply from 172.36.0.10: bytes=32 time=2ms TTL=124

Reply from 172.36.0.10: bytes=32 time=2ms TTL=124

Reply from 172.36.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 172.36.0.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\User>ping 192.50.20.100

Pinging 192.50.20.100 with 32 bytes of data:

Reply from 192.50.20.100: bytes=32 time<1ms TTL=128

Reply from 192.50.20.100: bytes=32 time<1ms TTL=128

Reply from 192.50.20.100: bytes=32 time<1ms TTL=128

Reply from 192.50.20.100: bytes=32 time<1ms TTL=128

Ping statistics for 192.50.20.100:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Documents and Settings\User>

!!!!!! N

192.50.20.100

Win-1

223.0.50.10

Win-2

172.36.0.10

Win-3

R5

Net

Physical

01 : 00 : 00

START

12:47 AM

7/20/2026

```
R6
R6#sh ip int brief
Interface      IP-Address      OK? Method Status      Prot
ocol
Ethernet0/0    223.0.50.1      YES manual up          up
Ethernet0/1    26.0.0.2        YES manual up          up
Ethernet0/2    unassigned      YES TFTP  down        down
Ethernet0/3    unassigned      YES TFTP  down        down

R6#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S 12.0.0.0/8 [1/0] via 26.0.0.1
S 23.0.0.0/8 [1/0] via 26.0.0.1
S 24.0.0.0/8 [1/0] via 26.0.0.1
S 26.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 26.0.0.0/8 is directly connected, Ethernet0/1
L 26.0.0.2/32 is directly connected, Ethernet0/1
S 32.0.0.0/8 [1/0] via 26.0.0.1
S 45.0.0.0/8 [1/0] via 26.0.0.1
S 172.36.0.0/16 [1/0] via 26.0.0.1
S 192.50.20.0/24 [1/0] via 26.0.0.1
S 223.0.50.0/24 is variably subnetted, 2 subnets, 2 masks
C 223.0.50.0/24 is directly connected, Ethernet0/0
L 223.0.50.1/32 is directly connected, Ethernet0/0
R6#
R6#
```



```
R5
R5#sh ip int brief
Interface      IP-Address      OK? Method Status      Prot
ooc0l
Ethernet0/0     172.36.0.1      YES manual up          up
Ethernet0/1     32.0.0.2        YES manual up          up
Ethernet0/2     45.0.0.2        YES manual up          up
Ethernet0/3     unassigned      YES TFTP  down        down

R5#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    12.0.0.0/8 [1/0] via 45.0.0.1
      [1/0] via 32.0.0.1
S    23.0.0.0/8 [1/0] via 45.0.0.1
      [1/0] via 32.0.0.1
S    24.0.0.0/8 [1/0] via 45.0.0.1
      [1/0] via 32.0.0.1
S    26.0.0.0/8 [1/0] via 45.0.0.1
      [1/0] via 32.0.0.1
      32.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    32.0.0.0/8 is directly connected, Ethernet0/1
L    32.0.0.2/32 is directly connected, Ethernet0/1
C    45.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    45.0.0.0/8 is directly connected, Ethernet0/2
L    45.0.0.2/32 is directly connected, Ethernet0/2
      172.36.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.36.0.0/16 is directly connected, Ethernet0/0
L    172.36.0.1/32 is directly connected, Ethernet0/0
S    192.50.20.0/24 [1/0] via 45.0.0.1
      [1/0] via 32.0.0.1
S    223.0.50.0/24 [1/0] via 45.0.0.1
      [1/0] via 32.0.0.1
R5#
R5#
```

```
R4
R4(config)#
R4#s
*Jul 19 19:13:30.350: %SYS-5-CONFIG_I: Configured from console by console
R4#sh ip int brief
Interface                IP-Address      OK? Method Status      Protocol
Ethernet0/0              24.0.0.2        YES manual up          up
Ethernet0/1              45.0.0.1        YES manual up          up
Ethernet0/2              unassigned      YES TFTP  down        down
Ethernet0/3              unassigned      YES TFTP  down        down
R4#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    12.0.0.0/8 [1/0] via 24.0.0.1
S    23.0.0.0/8 [1/0] via 24.0.0.1
    24.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    24.0.0.0/8 is directly connected, Ethernet0/0
L    24.0.0.2/32 is directly connected, Ethernet0/0
S    26.0.0.0/8 [1/0] via 24.0.0.1
S    32.0.0.0/8 [1/0] via 45.0.0.2
        [1/0] via 24.0.0.1
    45.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    45.0.0.0/8 is directly connected, Ethernet0/1
L    45.0.0.1/32 is directly connected, Ethernet0/1
S    172.36.0.0/16 [1/0] via 45.0.0.2
S    192.50.20.0/24 [1/0] via 24.0.0.1
S    223.0.50.0/24 [1/0] via 24.0.0.1
R4#
R4#
```

```
R3
R3(config)#
R3#
*Jul 19 19:12:47.790: %SYS-5-CONFIG_I: Configured from console by console
R3#sh ip int brief
Interface                IP-Address      OK? Method Status      Protocol
Ethernet0/0              23.0.0.2        YES manual up          up
Ethernet0/1              32.0.0.1        YES manual up          up
Ethernet0/2              unassigned      YES TFTP  down        down
Ethernet0/3              unassigned      YES TFTP  down        down
R3#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    12.0.0.0/8 [1/0] via 32.0.0.2
    23.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    23.0.0.0/8 is directly connected, Ethernet0/0
L    23.0.0.2/32 is directly connected, Ethernet0/0
S    24.0.0.0/8 [1/0] via 23.0.0.1
    26.0.0.0/8 [1/0] via 23.0.0.1
    32.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    32.0.0.0/8 is directly connected, Ethernet0/1
L    32.0.0.1/32 is directly connected, Ethernet0/1
S    45.0.0.0/8 [1/0] via 23.0.0.1
S    172.36.0.0/16 [1/0] via 32.0.0.2
S    192.50.20.0/24 [1/0] via 23.0.0.1
S    223.0.50.0/24 [1/0] via 23.0.0.1
R3#
R3#
```



Search

12:43 AM
7/20/2026


```
R2
R2#sh ip int brief
Interface                IP-Address      OK? Method Status        Protocol
Ethernet0/0              12.0.0.2        YES manual up             up
Ethernet0/1              26.0.0.1        YES manual up             up
Ethernet0/2              23.0.0.1        YES manual up             up
Ethernet0/3              24.0.0.1        YES manual up             up
R2#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

 12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    12.0.0.0/8 is directly connected, Ethernet0/0
L    12.0.0.2/32 is directly connected, Ethernet0/0
C    23.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    23.0.0.0/8 is directly connected, Ethernet0/2
L    23.0.0.1/32 is directly connected, Ethernet0/2
C    24.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    24.0.0.0/8 is directly connected, Ethernet0/3
L    24.0.0.1/32 is directly connected, Ethernet0/3
C    26.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    26.0.0.0/8 is directly connected, Ethernet0/1
L    26.0.0.1/32 is directly connected, Ethernet0/1
S    32.0.0.0/8 [1/0] via 23.0.0.2
S    45.0.0.0/8 [1/0] via 24.0.0.2
S    172.36.0.0/16 [1/0] via 24.0.0.2
                        [1/0] via 23.0.0.2
S    192.50.20.0/24 [1/0] via 12.0.0.1
S    223.0.50.0/24 [1/0] via 26.0.0.2
R2#
```

```
R1
R1#sh ip int brief
Interface      IP-Address      OK? Method Status      Prot
ooc1
Ethernet0/0     192.50.20.1     YES manual up          up
Ethernet0/1     12.0.0.1        YES manual up          up
Ethernet0/2     unassigned      YES TFTP  down        down
Ethernet0/3     unassigned      YES TFTP  down        down

R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

    12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       12.0.0.0/8 is directly connected, Ethernet0/1
L       12.0.0.1/32 is directly connected, Ethernet0/1
S       23.0.0.0/8 [1/0] via 12.0.0.2
S       24.0.0.0/8 [1/0] via 12.0.0.2
S       26.0.0.0/8 [1/0] via 12.0.0.2
S       32.0.0.0/8 [1/0] via 12.0.0.2
S       45.0.0.0/8 [1/0] via 12.0.0.2
S       172.36.0.0/16 [1/0] via 12.0.0.2
    192.50.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.50.20.0/24 is directly connected, Ethernet0/0
L       192.50.20.1/32 is directly connected, Ethernet0/0
S       223.0.50.0/24 [1/0] via 12.0.0.2

R1#
```

(116) Resume Building Session

Check Telnet SSH Status

PNETLab | Topology

(125) YouTube

Ask Gemini

labs.surrennetwork.com/legacy/topology

QEMU (Win-1)

!!!!!! N

```
C:\WINDOWS\system32\cmd.exe
Microsoft Windows [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\User>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 192.50.20.100
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.50.20.1

C:\Documents and Settings\User>
```

Win-1

192.50.20.100

Win-2

223.0.50.10

Win-3

172.36.0.10

Physical

01 : 00 : 00

START

11:20 PM

7/19/2026

QEMU (Win-2)

```
C:\WINDOWS\system32\cmd.exe
Microsoft Windows [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\User>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 223.0.50.10
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 223.0.50.1

C:\Documents and Settings\User>
```

!!!!!! N

192.50.20.100

Win-1

Win-2

223.0.50.10

Win-3

172.36.0.10

Physical

01 : 00 : 00

START

11:19 PM
7/19/2026